

No. of Functional Features Included in the Solution

Date	18 July 2026
Team ID	XXXXXX
Project Name	Smart Lender: Loan Eligibility Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Single Applicant Prediction	Allows user to fill a form with profile, income, credit history, and loan details to get real-time eligibility status.	Application Module	Done	1.0
2	Batch Prediction (CSV Upload)	Supports bulk processing of loan profiles by uploading a CSV file and evaluating them together.	Batch Prediction	Done	1.0
3	Data Preprocessing Pipeline	Handles automated data cleaning, missing value resolution, categorical encoding, and feature transformation.	Technical Architecture	Done	0.5
4	ML Model Training Suite	Trains and compares multiple classification models including Decision Tree, Random Forest, KNN, and XGBoost.	Technical Architecture	Done	0.75

5	Model Evaluation & Selection	Evaluates models based on Accuracy, Precision, Recall, F1-Score, and ROC-AUC to save the best serialized model file.	Technical Architecture	Done	0.5
6	Flask Web Integration	Hosts the predictive pipeline inside a Flask web application, serving interactive user forms and results.	Technical Architecture	Done	
7	History Tracking & Database	Stores every runtime applicant prediction and confidence score securely into a history database for auditing.	Technical Architecture	Done	0.5
8	Analytical Dashboards & Reports	Generates downloadable summary reports and displays visual trends of loan approval rates and model metrics.	Technical Architecture	Done	0.25

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Single Applicant Form, File Uploader Dashboard	Web Application Input Forms
2	Backend / Logic	ML Pipeline Pipeline, Model Prediction Engine	Decision Tree / Random Forest Inference
3	Database / Storage	Applicant Logs Database, Historical Storage	History tracking SQL/NoSQL engine
4	API / Integration	Flask Endpoint Layer, Batch processing hooks	Real-time JSON endpoint calls
5	Security / Authentication	Input Verification, Form Access Control	Data Validation & Model File Protection